

Hamstring Tendon Versus Bone-Patellar Tendon-Bone Autograft for ACL Reconstruction with Concurrent Lateral Extra-Articular Procedure

A Systematic Review and Network Meta-Analysis of Randomized Controlled Trials

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Background: No studies have yet evaluated the clinical outcomes of different anterior cruciate ligament (ACL) autografts in combination with a lateral extra-articular procedure (LEP). Thus, we systematically reviewed randomized controlled trials (RCTs) and conducted a network meta-analysis (NMA) to compare graft options for ACL reconstruction (ACLR) with concurrent LEP.

Methods: A systematic search in the PubMed, Embase, Cochrane Library, and Google Scholar databases identified RCTs on primary ACLR plus LEP. Data on ACLR failure, residual knee instability, patient-reported outcome measures, and complications were analyzed using NMA.

Results: On the basis of 13 studies (1,690 patients), ACLR with a hamstring tendon (HT) autograft plus LEP was associated with significantly lower odds of graft rupture (odds ratio [OR]: 0.28; 95% confidence interval [CI]: 0.16 to 0.50), graft failure (OR: 0.27; 95% CI: 0.15 to 0.47), clinical failure (OR: 0.48; 95% CI: 0.36 to 0.65), and residual pivot shift (OR: 0.43; 95% CI: 0.22 to 0.84) compared with isolated ACLR. Bone-patellar tendon-bone (BPTB) autograft plus LEP was associated with a significantly lower clinical failure rate (OR: 0.30; 95% CI: 0.12 to 0.80) compared with isolated ACLR. Both HT (mean difference [MD]: 2.40; 95% CI: 1.25 to 3.55) and BPTB (MD: 3.70; 95% CI: 0.85 to 6.55) autograft plus LEP were associated with higher Lysholm scores compared with isolated ACLR. The outcomes did not differ between the graft types when combined with LEP.

Conclusions: The odds of graft rupture, graft failure, and clinical failure after ACLR with HT autograft plus LEP were lower by 72%, 73%, and 52%, respectively, than the odds after isolated ACLR. ACLR with BPTB autograft plus LEP significantly lowered only the odds of clinical failure, by approximately 70%, potentially due to the smaller sample size. Both grafts remain viable options for ACLR plus LEP, with the benefit of LEP requiring further validation.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

Graft selection in anterior cruciate ligament (ACL) reconstruction (ACLR) is critical, as it affects clinical outcomes^{1,2}. The graft failure rate after primary ACLR is approximately 7 times higher for allografts than for bone-patellar tendon-bone (BPTB) and hamstring tendon (HT) autografts³⁻⁶. However, the risk of revision is twice as high for HT autografts than for BPTB or quadriceps tendon autografts⁷⁻⁹.

Graft failure and residual rotational instability remain important concerns, with rates of up to 14% and 30%, respectively¹⁰⁻¹². Lateral extra-articular procedures (LEPs), such as anterolateral ligament reconstruction (ALLR) and lateral extra-articular tenodesis (LET), address these issues by providing secondary stabilization, mitigating the pivot shift phenomenon¹³⁻¹⁵. Recent meta-analyses showed that combining ACLR

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with LEP reduces graft failure and rotational instability more effectively than ACLR alone¹⁶⁻²¹.

Despite the importance of graft choice and the success of ACLR plus LEP, the optimal graft for combined ACLR and LEP remains unknown. No previous studies have compared different types of autografts used for ACLR with concurrent LEP. Therefore, we conducted a systematic review and network meta-analysis (NMA) to evaluate failure rates and outcomes according to graft type in ACLR plus LEP, hypothesizing superiority of BPTB over HT.

Materials and Methods

Search Strategy

The review was registered in PROSPERO (CRD42023485705) and conducted according to the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) guidelines. A systematic search was performed in the PubMed, Embase, Cochrane Library, and Google Scholar databases for articles published through November 21, 2024, using the following terms: (“Anterior Cruciate Ligament” [MeSH] OR “Anterior cruciate ligament” OR

“ACL”) AND (“Tenodesis” OR “Lateral plasty” OR “Extra articular” OR “Anterolateral ligament” OR “ALL”).

Eligibility

The study inclusion criteria were (1) skeletally mature individuals with an ACL rupture, (2) primary single-bundle ACLR plus LET or ALLR, (3) a Level-I randomized controlled trial (RCT) in English, (4) ≥12-month follow-up, (5) specified graft types, and (6) reported outcomes, such as graft failure rate, revision rate, and/or patient-reported outcome measures (PROMs). The exclusion criteria were studies (1) with incomplete data, (2) of revision ACLR, and (3) with Level-II to V evidence. Two independent reviewers screened the results; disagreements were resolved through discussion.

Data Extraction

Two reviewers collected the data, including the study details (author, year, country, design, sample size, follow-up), patient characteristics (sex, age, body mass index), and outcomes (failure rate, residual pivot shift, anteroposterior laxity, International

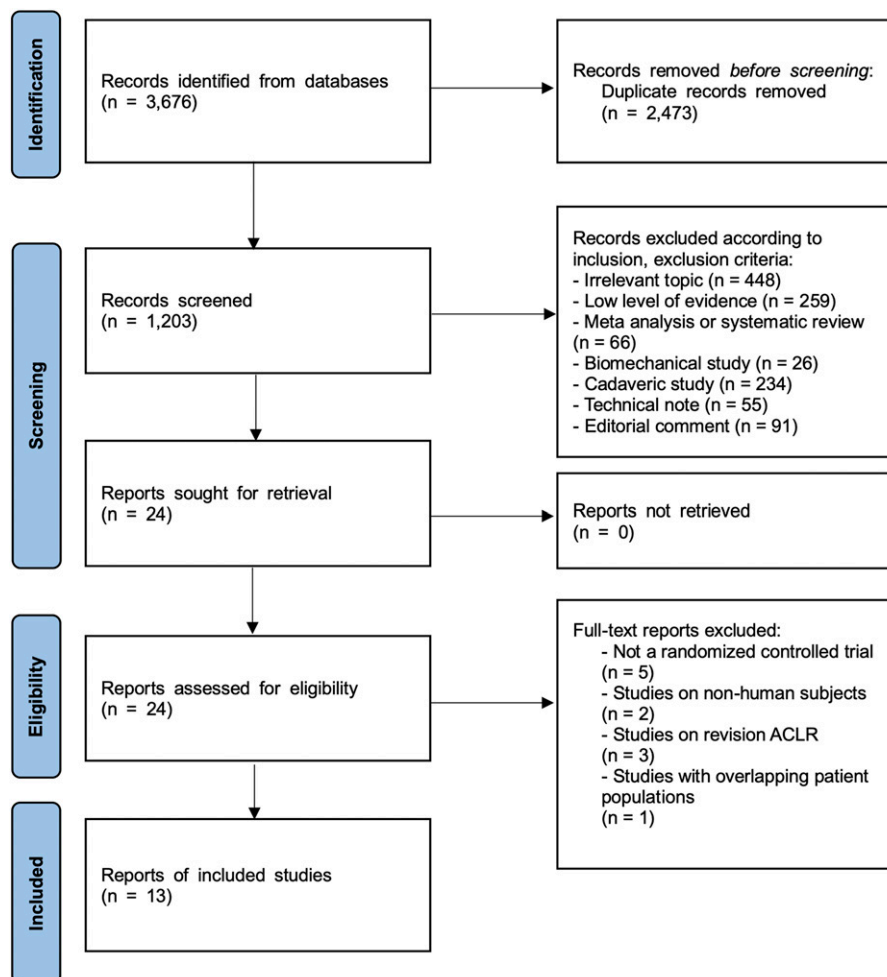


Fig. 1

PRISMA flow diagram for the systematic review. ACLR = anterior cruciate ligament reconstruction.

Knee Documentation Committee [IKDC] score, Lysholm score, Tegner score, complications). Missing or unclear data were clarified by contacting the authors. The risk of bias was assessed using the Cochrane Collaboration's risk-of-bias tool (RoB 2), version 2²². Disagreements were adjudicated by a senior reviewer.

Statistical Analysis

All statistical analyses and visualizations were performed using R software (version 4.2.1; R Foundation for Statistical Computing), with significance set at $p < 0.05$. Means and standard deviations of numerical variables were recorded, with missing standard deviations estimated using the method proposed by Furukawa et al.²³. For graphical data without numerical values, extraction was performed using the approach described by Gheibi et al.²⁴. To avoid patient overlap among studies by the same group, only the largest study from the group was included in the meta-analysis; however, the others were retained in the review.

Continuous outcomes were analyzed using mean differences (MDs) with 95% confidence intervals (CIs), and dichotomous outcomes were analyzed using odds ratios (ORs) with 95% CIs. Heterogeneity was assessed using the I^2 statistic, with substantial heterogeneity defined as $I^2 > 50\%$; a pooled synthesis was not conducted in such cases. Additionally, subgroup analyses were performed on the basis of the graft type used in isolated ACLR and the LEP type to explore potential sources of heterogeneity.

An NMA compared the treatment effects of HT and BPTB autografts for ACLR plus LEP versus isolated ACLR (control). Outcomes included ACLR failure, residual pivot shift, and the Lysholm score. ACLR failure was categorized as graft rupture, graft failure (rupture or revision), or clinical failure (rupture, revision, pivot shift grade of 2 or 3, or residual

anteroposterior laxity of >5 mm). When clinical failure components overlapped within a study, the highest reported number was used for analysis. Separate NMAs were performed for each failure definition.

NMA was conducted using the netmeta package (version 2.9-2) in R. Transitivity—a key NMA assumption requiring similarity in effect modifiers for valid comparisons²⁵—was evaluated qualitatively and via meta-regression on patient age, sex, and inclusion of high-risk individuals. A global consistency test and a node-splitting analysis were not feasible due to the absence of closed loops.

The surface under the cumulative ranking curve (SU-CRA) was calculated to rank treatments from most to least effective. SUCRA values (0 to 1) represent the cumulative probability distribution of treatment rankings, with higher values indicating greater efficacy.

Results

Study Characteristics

Thirteen studies comprising 1,690 patients were included in the final analysis (Fig. 1). Tables I and II present the detailed study characteristics, and Appendix Supplementary Table 1 summarizes concomitant injuries and surgeries.

Methodological Quality Assessment

The methodological quality of the included Level-I RCTs was assessed and presented as a risk-of-bias summary (Fig. 2-A) and a traffic light plot (Fig. 2-B). Qualitatively, the studies included in the NMA were comparable across key effect modifiers. Additionally, meta-regression analyses showed no significant association of these modifiers with any outcomes (see Appendix Supplementary Fig. 1), supporting the transitivity assumption. Subgroup analyses for each outcome showed no

TABLE I Overview of the Included Randomized Controlled Trials*

Study	Year	Country	No. of Patients	Graft Used for ACLR	Type of Lateral Procedure
Mogoş et al. ²⁶	2023	Italy	57	Autologous HT	ALLR (autologous HT)
El-Azab et al. ³⁴	2023	Egypt	100	Autologous HT	LET
Grassi et al. ²⁷	2022	Italy	59	Autologous HT	LET
Hamido et al. ³⁶	2021	Kuwait	107	Autologous HT	ALLR (autologous HT)
Castoldi et al. ³¹	2020	France	120	Autologous BPTB	LET
Getgood et al. ³⁰	2020	Canada	618	Autologous HT	LET
Porter and Shadbolt ²⁹	2020	Australia	55	Autologous HT	LET
Sonnery-Cottet et al. ²⁸	2020	France	224	Autologous HT	ALLR (autologous HT)
Ibrahim et al. ³⁸	2017	Kuwait	110	Autologous HT	ALLR (autologous HT)
Trichine et al. ³⁵	2014	Algeria	120	Autologous BPTB	LET
Vadalà et al. ³²	2013	Italy	60	Autologous HT	LET
Zaffagnini et al. ³⁷	2006	Italy	75	Autologous HT	LET
Anderson et al. ³³	2001	United States	105	Autologous HT	LET

*ACLR = anterior cruciate ligament reconstruction, HT = hamstring tendon, ALLR = anterolateral ligament reconstruction, LET = lateral extra-articular tenodesis, BPTB = bone-patellar tendon-bone.

TABLE II Patient Demographics and Inclusion and Exclusion Criteria Used in the Included Studies*

Graft Used for ACLR	Study	Follow-up Duration†	Sex, M:F‡	Age†† (yr)	BMI†† (kg/m ²)	Inclusion Criteria	Exclusion Criteria
Autologous HT	Mogoş et al., 2023 ²⁶	12 mo	17:8/27:5	31 ± 7.1	NR	(1) Unilateral primary ACL tear (2) Participation in high-grade pivoting sports (3) Preoperative pivot-shift test result showing grade-2 or 3 instability	(1) Recurrent ACL tear (2) Previous knee surgery (3) Presence of multiligament knee injuries (4) Skeletal immaturity (5) Requirement for additional extra-articular procedures (except ALL reconstruction) or PCL reconstruction (6) Diagnosis of rheumatoid arthritis
	El-Azab et al., 2023 ³⁴	24 mo	41:7/39:8	27 ± 5.85/ 28 ± 6.0	NR	(1) ACL-deficient knee confirmed by physical examination and MRI (2) Patient age of 18-45 years (3) Skeletal maturity (4) Pivot-shift test result showing grade-2 or 3 instability	(1) History of revision surgery or ACL tear in the contralateral knee (2) Presence of multiligament knee injuries or polytrauma (3) Generalized ligamentous laxity (4) Symptomatic articular cartilage defects requiring treatment (Outerbridge grade of >II). (5) Varus or valgus malalignment exceeding 3°
	Grassi et al., 2022 ²⁷	25.0 ± 5.6 mo	13:2/20:4	23.4 ± 4.7/ 24.1 ± 6.8	NR	(1) ACL deficiency (2) Patient age of 18-65 years	(1) Previous knee surgery (2) Presence of concomitant ligamentous injuries
	Hamido et al., 2021 ³⁶	Median, 60 mo (range, 55-65 mo)	52:0/50:0	26 (18-40)/ 24 (18-33)	NR	(1) Combined ACL and ALL tears and grade-3 pivot shift in a male athlete (2) Presence of a Segond fracture (3) Participation in high-level sports or sports involving frequent pivoting movements	(1) Presentation for revision ACLR surgery (2) Previous knee surgery or knee dislocation (3) Presence of multiligament knee injuries (4) Presence of osteoarthritis
	Getgood et al., 2020 ³⁸	2 yr	151:161/ 151:155	18.8 ± 3.2/ 19.1 ± 3.3	23.8 ± 3.7/ 24 ± 3.8	(1) ACL deficiency (2) Patient age of 14-25 years (3) Considered at higher risk of reinjury based on the presence of at least 2 of the following factors: (a) participation in competitive pivoting sports, (b) preoperative pivot-shift test result showing grade-2 or 3 instability, and/or (c) generalized ligamentous laxity (Beighton score of ≥4) or genu recurvatum of >10°	(1) Previous ACLR of either knee (2) Presence of multiligament knee injuries (3) Skeletal immaturity (4) Symptomatic articular cartilage defects requiring treatment (5) Varus malalignment exceeding 3° (6) Patient inability or unwillingness to comply with a 2-year postoperative follow-up
	Porter and Shadbolt, 2020 ³⁹	2 yr	15:12/17:11	22.3 ± 3.7/ 21.8 ± 4.1	21.1 ± 1.8/ 21 ± 1.5	(1) ACL rupture confirmed by MRI and arthroscopic surgery (2) Skeletal maturity (3) Non-contact ACL injury sustained during twisting or pivoting sports (4) ACLR performed within 6 weeks after injury (5) Pivot-shift test result at least 1 instability grade higher compared with the contralateral knee after ACLR	(1) Previous ACL injury in either knee (2) Presence of other ligament injuries with grade of >1 or reparable meniscal tears (3) Diagnosis of rheumatoid arthritis, connective tissue disorder, or autoimmune disease (4) Unwillingness to be randomized into a treatment group (5) ACL injury sustained >6 weeks prior to surgery (6) Patient not fit for general anesthesia
	Sonnery-Cottet et al., 2020 ³⁸	12.3 ± 1.9 mo	96:16/78:34	25.9 ± 4.7/ 24.7 ± 4.4	NR	(1) ACL deficiency requiring reconstruction (2) Patient age of 18-35 years	(1) Previous ligament surgery in either knee (2) Presence of multiligament knee injuries (3) BMI of <18.5 or >30 kg/m ² (4) Professional athlete (5) Articular cartilage defects requiring treatment (6) Connective tissue disorder or congenital disease (7) Patient refusal to participate in the study
	Ibrahim et al., 2017 ³⁸	27 mo	50:0/53:0	26 (20-30)/ 26 (21-32)	NR	(1) Unilateral primary ACL tear in a male athlete	(1) Presentation for revision ACLR (2) Multiligament knee injuries
	Vadalà et al., 2013 ³⁷	44.6 mo	0:28/0:27	27 (15-40)	NR	(1) Surgery for chronic ACL instability in a female patient (2) Patient age of <40 years (3) Preoperative pivot-shift test result showing grade-2 or 3 instability (4) Minimum of 2 months between trauma and surgery	(1) Previous ligament surgery in either knee (2) Presence of osteoarthritis (3) Imaging evidence of grade-III or IV chondral damage on both patellar surfaces or medial and lateral femoral condyles, according to the International Cartilage Repair Society (4) Presence of concomitant systemic disease
	Zaffagnini et al., 2006 ³⁷	5 yr	15:10/18:7	31.3 (26-49)/26.7 (15-44)	NR	(1) ACL rupture (2) Patient age of <50 years (3) Intact PCL and maximum grade-1 MCL injury (4) Patient involved in stopping sports at the competitive or amateur level	(1) Previous knee surgery (2) Presence of meniscal injury (3) Abnormal contralateral knee (4) Presence of articular cartilage injury (5) Presence of osteoarthritis (6) History of patellofemoral pain
	Anderson et al., 2001 ³⁸	35.4 ± 11.6 mo	46:24/22:13	21.9 (14-44)/22 (14-40)	NR	(1) Unilateral ACL tear (2) Patient age of <50 years (3) Patient willingness to be subjected to randomization (4) Normal contralateral knee (5) Absence of substantial injury to other ligamentous structures (less than grade-1 injury to the MCL)	(1) Previous knee surgery other than meniscectomy (2) Presence of osteoarthritis in either knee (3) History of patellofemoral pain or considerable patellofemoral crepitation

continued

TABLE II (continued)

Graft Used for ACLR	Study	Follow-up Duration†	Sex, M:F‡	Age†§ (yr)	BMI†§ (kg/m ²)	Inclusion Criteria	Exclusion Criteria
Autologous BPTB	Castoldi et al., 2020 ³¹	19.4 yr	43:18/47:13	25.9 (16-40)/26.2 (15-40)	NR	(1) Complete isolated primary ACL rupture with a plan for arthroscopic ACLR	(1) History of ACL repair or reconstruction (2) Presence of multiligament knee injuries
	Trichine et al., 2014 ³²	24.5 mo	60:0/60:0	28.6 ± 4.7/27.7 ± 4.75	NR	(1) ACL rupture with advanced chronic anterior laxity (defined as a positive Lachman test result [graded 3+], grade 3+ pivot shift, and a positive anterior drawer test result at 90° of flexion [graded 2+ or 3+]) (2) Patient willingness to be subjected to randomization	(1) Previous knee surgery (2) Presence of multiligament knee injuries (3) Injury to the contralateral knee (4) Skeletal immaturity

*ACL = anterior cruciate ligament, ACLR = ACL reconstruction, BMI = body mass index, HT = hamstring tendon, NR = not reported, ALL = anterolateral ligament, MRI = magnetic resonance imaging, MCL = medial collateral ligament, PCL = posterior cruciate ligament, BPTB = bone-patellar tendon-bone. †The values are given as the mean, the mean ± standard deviation, or the median (range).
‡For isolated ACLR/for ACLR + lateral extra-articular procedure, or overall. Numbers of patients may be less than in Table I due to loss to follow-up.

significant differences according to graft type in isolated ACLR or according to LEP type (see Appendix Supplementary Figs. 2 and 3).

Graft Rupture

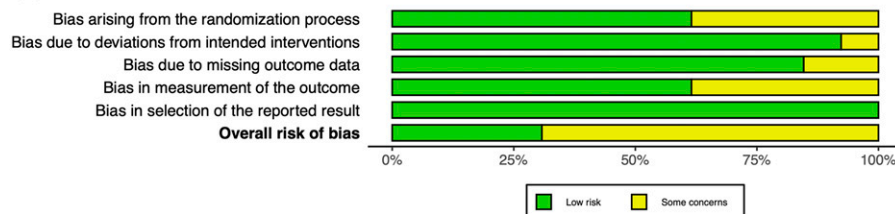
The NMA on graft rupture included 9 studies (Fig. 3-A)²⁶⁻³⁴. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.98$). Adding LEP to ACLR with HT autograft significantly reduced the odds of graft rupture (OR: 0.28; 95% CI: 0.16 to 0.50) compared with isolated ACLR, while the effect of adding LEP to ACLR with BPTB

failed to reach significance (OR: 0.26; 95% CI: 0.03 to 2.41). No difference was observed between graft types in ACLR plus LEP. The SUCRA rankings placed ACLR plus LEP higher (HT: 0.73; BPTB: 0.72) than isolated ACLR (0.05).

Graft Failure

Eleven studies reported graft failure, including rupture or revision surgery (Fig. 3-B)²⁶⁻³⁶. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.98$). Adding LEP to ACLR with HT autograft significantly reduced the odds of graft failure (OR: 0.27; 95% CI:

(a) Summary of RoB 2.0 assessments



(b) Traffic light plot of RoB 2.0 assessments

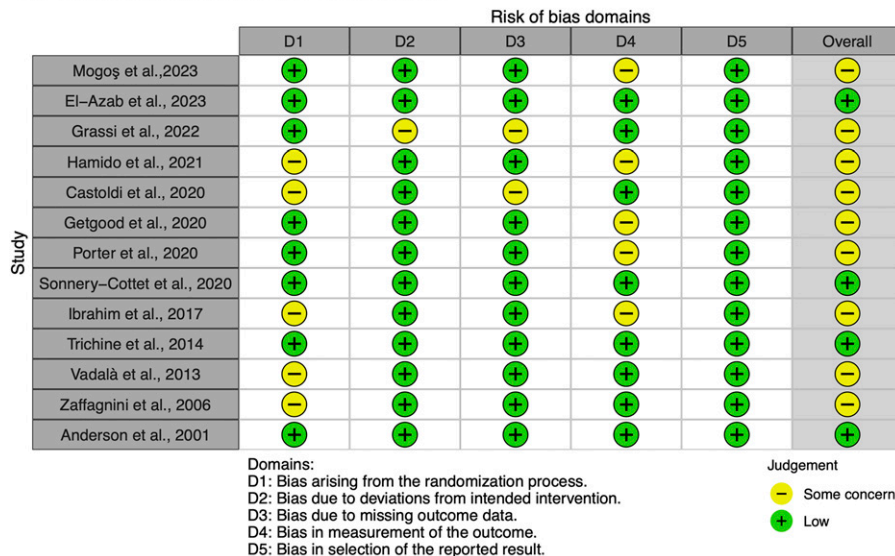


Fig. 2
Risk-of-bias (RoB) assessments for the included studies. **Fig. 2-A** Summary of the RoB 2.0 results. **Fig. 2-B** Traffic light plot illustrating the RoB 2.0 results.

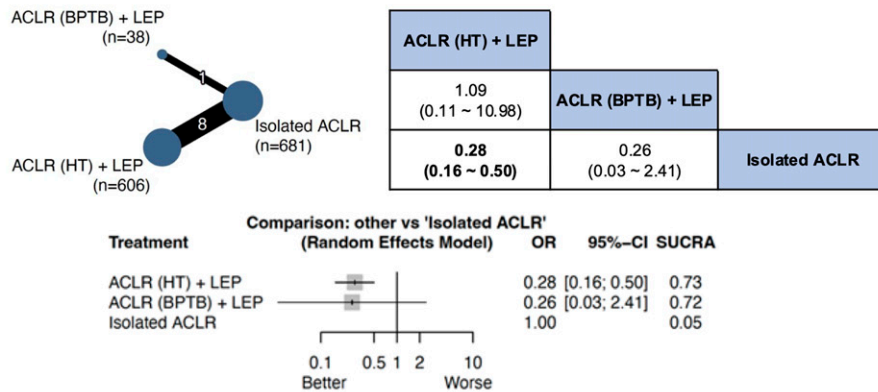
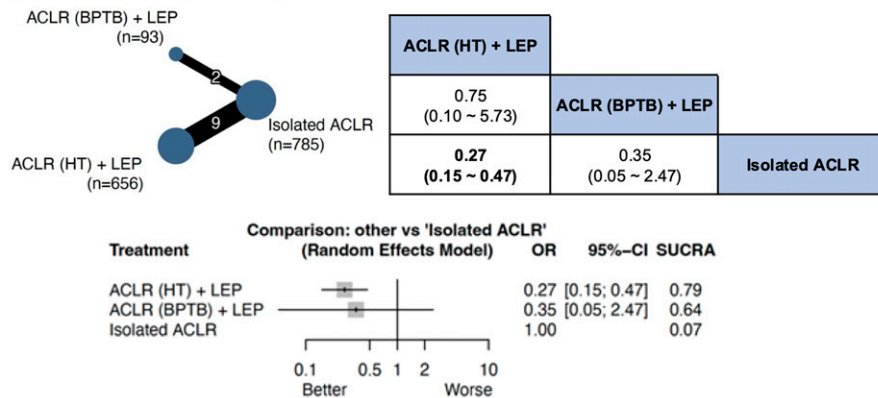
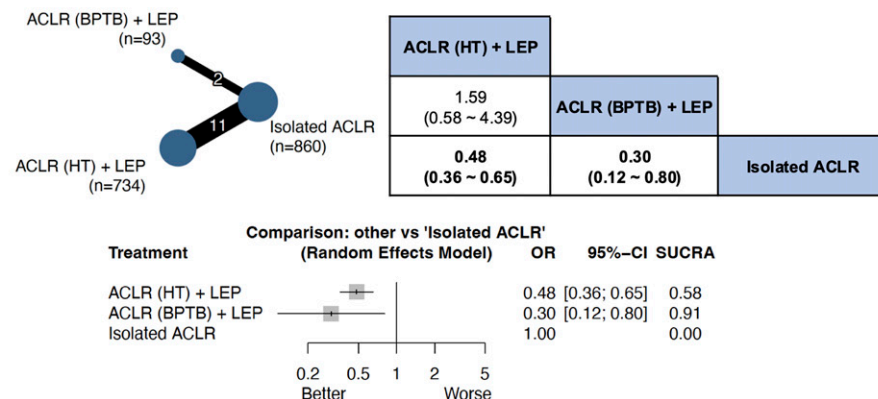
(a) Graft rupture ($I^2 = 0\%$)**(b) Graft failure ($I^2 = 0\%$)****(c) Clinical failure ($I^2 = 0\%$)**

Fig. 3

Edge plots, league tables, and forest plots for graft rupture (**Fig. 3-A**), graft failure (**Fig. 3-B**), and clinical failure (**Fig. 3-C**). The I^2 statistic represents heterogeneity. In the edge plot, the width of each line is proportional to the number of trials comparing the pair of treatments, the number on the line represents the number of these trials, and the size of each circle is proportional to the number of cases. The league plot shows the standardized mean difference and 95% confidence interval for each treatment pair; significant findings are in bold. ACLR = anterior cruciate ligament reconstruction, HT = hamstring tendon, BPTB = bone-patellar tendon-bone, LEP = lateral extra-articular procedure, OR = odds ratio, CI = confidence interval, SUCRA = surface under the cumulative ranking curve.

0.15 to 0.47) compared with isolated ACLR, while the effect of adding LEP to ACLR with BPTB failed to reach significance (OR: 0.35; 95% CI: 0.05 to 2.47). No difference was observed between

graft types in ACLR plus LEP. The SUCRA rankings placed ACLR with HT autograft plus LEP first (0.79), followed by BPTB autograft plus LEP (0.64) and isolated ACLR (0.07).

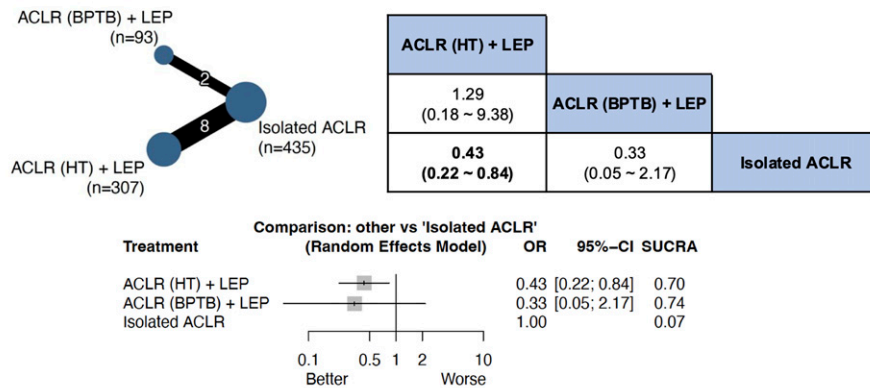
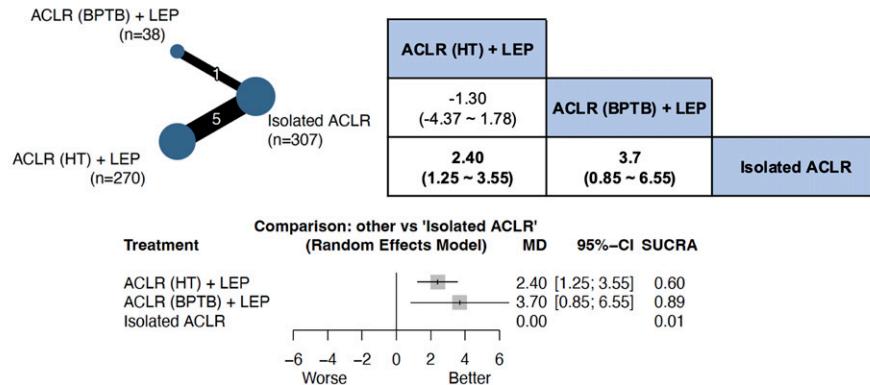
(a) Residual pivot shift ($I^2 = 0\%$)**(b) Lysholm score ($I^2 = 0\%$)**

Fig. 4
Edge plots, league tables, and forest plots for residual pivot shift (Fig. 4-A) and Lysholm score (Fig. 4-B). The I^2 statistic represents heterogeneity. In the edge plot, the width of each line is proportional to the number of trials comparing the pair of treatments, the number on the line represents the number of these trials, and the size of each circle is proportional to the number of cases. The league plot shows the standardized mean difference and 95% confidence interval for each treatment pair; significant findings are in bold. ACLR = anterior cruciate ligament reconstruction, HT = hamstring tendon, BPTB = bone-patellar tendon-bone, LEP = lateral extra-articular procedure, OR = odds ratio, MD = mean difference, CI = confidence interval, SUCRA = surface under the cumulative ranking.

Clinical Failure

The NMA on clinical failure—defined as graft rupture, revision surgery, grade-2 or 3 pivot shift, or residual anteroposterior laxity of >5 mm—included 13 studies (Fig. 3-C)²⁶⁻³⁸. Clinical failure was defined by pivot shift in 7 of these studies^{29,30,32-36}, by residual anteroposterior laxity in 3^{31,37,38}, and by graft failure in 3²⁶⁻²⁸. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.79$). Adding LEP to ACLR with BPTB (OR: 0.30; 95% CI: 0.12 to 0.80) or with HT (OR: 0.48; 95% CI: 0.36 to 0.65) significantly reduced the odds of clinical failure compared with isolated ACLR. No differences were found between graft types in ACLR plus LEP. The SUCRA rankings placed BPTB autograft plus LEP first (0.91), followed by HT autograft plus LEP (0.58) and isolated ACLR (0.00).

Residual Pivot Shift

Ten studies reported residual pivot shift (grade 2 or 3) (Fig. 4-A)^{26,29,31-38}. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.58$). ACLR with HT autograft plus LEP significantly reduced the odds of a residual pivot shift (OR: 0.43; 95% CI: 0.22 to 0.84) compared with isolated ACLR, while the effect of adding LEP to

BPTB failed to reach significance (OR: 0.33; 95% CI: 0.05 to 2.17). No differences were found between graft types in ACLR plus LEP. The SUCRA rankings placed BPTB autograft plus LEP first (0.74), followed by HT autograft plus LEP (0.70) and isolated ACLR (0.07).

Lysholm Score

Six studies reported the Lysholm score (Fig. 4-B)^{28,29,31,32,36,38}. Low heterogeneity was observed ($I^2 = 0\%$, $\tau^2 = 0.0$, $p = 0.40$). Adding LEP was associated with significantly higher Lysholm scores compared with isolated ACLR for both BPTB (MD: 3.70; 95% CI: 0.85 to 6.55) and HT (MD: 2.40; 95% CI: 1.25 to 3.55). No differences were found between graft types in ACLR plus LEP. The SUCRA rankings placed BPTB autograft plus LEP first (0.89), followed by HT autograft plus LEP (0.60) and isolated ACLR (0.01).

Other Clinical Outcomes

Residual anteroposterior laxity and clinical scores (IKDC and Tegner activity) were extracted (Tables III and IV), but significant heterogeneity precluded pooled analysis.

TABLE III ACLR Failure, Residual Pivot Shift, and Residual Anteroposterior Laxity Reported in the Included Studies*

Graft Used for ACLR	Study	Graft Rupture (no. [%])		Graft Failure (no. [%])		Clinical Failure (no. [%])		Residual Pivot Shift (no. [%])		Residual AP Laxity (mm)		
		Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR†	ACLR + LEP†	Method†
Autologous HT	Mogoş et al., 2023 ³⁶	1 (4.0%)	0 (0%)	1 (4.0%)	0 (0%)	1 (4.0%)	0 (0%)	0 (0%)	0 (0%)	0.24 ± 0.43	0.13 ± 0.34	Rolimeter
	El-Azab et al., 2023 ³⁴	3 (6.3%)	1 (2.1%)	3 (6.3%)	1 (2.1%)	4 (8.4%)	1 (2.1%)	4 (8.3%)	1 (2.1%)	NR	NR	
	Grassi et al., 2022 ²⁷	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	NR	NR	1.5 ± 1.6	1.7 ± 2.2	KT-1000
	Hamido et al., 2021 ³⁶	NR	NR	0 (0%)	0 (0%)	5 (9.6%)	0 (0%)	5 (9.6%)	0 (0%)	2.5 ± 0.7	1.2 ± 0.7	KT-1000
	Getgood et al., 2020 ³⁰	34 (11.4%)	11 (3.8%)	34 (11.4%)	11 (3.8%)	120 (40.3%)	72 (24.7%)	NR	NR	NR	NR	
	Porter and Shadbolt, 2020 ²⁹	4 (17.4%)	0 (0%)	4 (17.4%)	0 (0%)	6 (26.1%)	4 (14.3%)	6 (26.1%)	4 (14.3%)	NR	NR	
	Sonnery-Cottet et al., 2020 ²⁸	6 (5.4%)	1 (0.9%)	6 (5.4%)	1 (0.9%)	6 (5.4%)	1 (0.9%)	NR	NR	1 ± 1	1 ± 1	Rolimeter
	Ibrahim et al., 2017 ³⁸	NR	NR	NR	NR	3 (6%)	1 (1.9%)	2 (4%)	1 (1.9%)	1.8 ± 0.8	1.3 ± 0.2	KT-1000
	Vadalà et al., 2013 ³²	2 (7.1%)	0 (0%)	2 (7.1%)	0 (0%)	4 (14.3%)	0 (0%)	4 (14.3%)	0 (0%)	2.8 ± 0.8	2.7 ± 0.9	KT-1000
	Zaffagnini et al., 2006 ³⁷	NR	NR	NR	NR	5 (20.0%)	3 (12.0%)	4 (16.0%)	0 (0%)	NR	NR	
Autologous BPTB	Anderson et al., 2001 ³³	2 (2.9%)	0 (0%)	2 (2.9%)	0 (0%)	8 (11.4%)	7 (20.0%)	8 (11.4%)	7 (20.0%)	3.1 ± 2.3	2.6 ± 2.2	KT-1000
	Castoldi et al., 2020 ³¹	4 (9.5%)	1 (2.6%)	4 (9.5%)	1 (2.6%)	16 (38.1%)	6 (15.8%)	1 (2.4%)	0 (0%)	NR	NR	
	Trichine et al., 2014 ³⁵	NR	NR	0 (0%)	0 (0%)	3 (5.8%)	1 (1.8%)	3 (5.8%)	1 (1.8%)	4.4 ± 4.8	4.1 ± 4.7	Telos

*AP = anteroposterior, ACLR = anterior cruciate ligament reconstruction, LEP = lateral extra-articular procedure, HT = hamstring tendon, NR = not reported, BPTB = bone-patellar tendon-bone. †The values are given as the mean ± standard deviation. ‡Rolimeter (Aircast), KT-100 (MEDmetric), or Telos (Telos).

Complications

Various complications were reported in studies comparing isolated ACLR to ACLR with HT autograft plus LEP. Hamido et al.³⁶ noted 2 cases of arthrofibrosis in their ALLR group, which resolved within 4 weeks with nonoperative management. Getgood et al.³⁰ reported more hardware irritation cases in their LET group (14) than in their isolated ACLR group (4). Sonnery-Cottet et al.²⁸ found a higher rate of reoperations for cyclops syndrome in their isolated ACLR group (10 of 112 versus 0 of 112, $p = 0.001$). Anderson et al.³³ reported additional procedures, such as meniscectomy or staple removal, after 6 procedures with LET and 5 with isolated ACLR. Porter and Shadbolt²⁹ noted similar meniscal tear rates in groups with and without LET, and no other complications. Ibrahim et al.³⁸ reported 1 wound infection in their ACLR plus ALLR group, whereas Mogoş et al.²⁶ found no complications in the ACLR plus ALLR group. Four studies^{27,32,34,37} reported no clinically important complications.

In a study comparing isolated ACLR to ACLR with BPTB autograft plus LEP, Trichine et al.³⁵ reported no complications or additional surgeries. However, Castoldi et al.³¹ observed higher lateral tibiofemoral osteoarthritis in their

ACLR with BPTB autograft plus LET group (59% versus 22%, $p = 0.02$) but no significant differences in medial or patellofemoral osteoarthritis.

Discussion

A key finding from our study is that adding LEP to ACLR with either HT or BPTB autograft was associated with significantly lower odds of ACLR failure compared with isolated ACLR. HT autografts plus LEP had 72%, 73%, and 52% lower odds of graft rupture, graft failure, and clinical failure, respectively. HT autografts also demonstrated superior rotational stability when used with LEP, as indicated by the pivot-shift test. The addition of LEP to ACLR with BPTB autograft yielded 70% lower odds of clinical failure; however, the impacts on graft rupture, graft failure, and the incidence of a pivot shift did not reach significance. These results highlight the need for additional studies to clarify the role of LEP when used with BPTB autografts. Notably, no significant differences were observed between HT and BPTB autografts when combined with LEP.

Graft rupture or revision surgery following isolated ACLR has been extensively studied, with the focus on choosing between

TABLE IV Clinical Outcome Scores Reported in the Included Studies*

Graft Used for ACLR	Study	Lysholm Score		IKDC Score		Tegner Score	
		Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP	Isolated ACLR	ACLR + LEP
Autologous HT	Mogoş et al., 2023 ²⁶	NR	NR	87 ± 9	91.5 ± 11	NR	NR
	El-Azab et al., 2023 ³⁴	NR	NR	87.9 ± 7.19	91.9 ± 8.9	NR	NR
	Grassi et al., 2022 ²⁷	NR	NR	NR	NR	NR	NR
	Hamido et al., 2021 ³⁶	94 ± 4.5	96 ± 5	NR	NR	7.8 ± 1.4	7.9 ± 0.8
	Getgood et al., 2020 ³⁰	NR	NR	86.6 ± 0.8	87.3 ± 0.8	NR	NR
	Porter and Shadbolt, 2020 ²⁹	92.5 ± 4.8	96.8 ± 8	90.9 ± 10.7	94.2 ± 11.2	7 ± 2.3	8 ± 1.9
	Sonnery-Cottet et al., 2020 ²⁸	88 ± 14	92 ± 10	81.2 ± 15.3	86.8 ± 12.3	6 ± 2	6 ± 2
	Ibrahim et al., 2017 ³⁸	96 ± 5.3	98 ± 7.6	NR	NR	8 ± 1.5	8 ± 1.5
	Vadalà et al., 2013 ³²	94.5 ± 6.7	95.8 ± 4	87 ± 1.8	89 ± 1.5	6.7 ± 1.4	7.5 ± 1.2
	Zaffagnini et al., 2006 ³⁷	NR	NR	76 ± 8	89 ± 13	7.1 ± 1.3	8.5 ± 1.9
	Anderson et al., 2001 ³³	NR	NR	NR	NR	NR	NR
Autologous BPTB	Castoldi et al., 2020 ³¹	86.6 ± 7.2	90.3 ± 5.8	81.1 ± 7.9	82.4 ± 6.7	NR	NR
	Trichine et al., 2014 ³⁵	NR	NR	92.1 ± 9.5	88.2 ± 8	NR	NR

*The values are given as the mean ± standard deviation. IKDC = International Knee Documentation Committee, ACLR = anterior cruciate ligament reconstruction, LEP = lateral extra-articular procedure, HT = hamstring tendon, NR = not reported, BPTB = bone-patellar tendon.

HT and BPTB grafts. Registry studies have reported a 1.6 to 6.1 times higher risk of graft rupture or revision with HT autografts compared with BPTB autografts^{9,39-42}, and a recent meta-analysis also found a 1.6 times higher risk⁴³. However, some other meta-analyses found no difference in the failure rates between the 2 grafts⁴⁴⁻⁴⁶. These discrepancies may result from study design heterogeneity and statistical fragility in ACLR research^{43,47}. Nevertheless, the significantly higher failure rates associated with HT autografts, or trends toward them, suggest that HT is not superior to BPTB in isolated ACLR.

Adding LEP to isolated ACLR seemingly mitigated or eliminated outcome differences between HT and BPTB autografts. The fact that, of the 3 types of failure assessed in this study, only clinical failure was found to have a significantly lower rate when LEP was added to the BPTB autografts may be attributable to their superior control of residual anteroposterior laxity, which is 1 of the criteria used to define clinical failure. BPTB autografts are reported to provide superior anteroposterior stability compared with HT autografts^{48,49}, and this inherent graft characteristic likely influenced our findings. Although the ACLR failure rate did not differ between the grafts when combined with LEP, these findings underscore the potential of LEP to enhance outcomes regardless of the graft type. However, these findings should be interpreted with caution, as factors such as age, sex, and sports participation may influence the outcomes. Maletis et al.⁹ reported that patients aged <21 years who underwent isolated ACLR with HT autografts had a 1.6 times higher revision risk than those with BPTB autografts. Salem et al.⁵⁰ also found higher graft rupture rates after isolated ACLR with HT autografts than with BPTB autografts in females aged 15 to 20 years (17.5% versus 6.4%). However, there is limited evidence regarding the comparative

outcomes of HT and BPTB autografts in combined ACLR and LEP, and we found none with risk stratification by age and sex. One study focused exclusively on high-risk patients aged <25 years³⁰, 1 included only females³², and 3 included only males^{35,36,38}. Thus, our findings reflect outcomes in a general ACL injury population rather than in specific age or risk groups. Further research examining the impacts of age and sex on graft outcomes in combined ACLR and LEP is needed.

Improved rotational stability, demonstrated by a decreased pivot shift phenomenon, is a key advantage of adding LEP to ACLR compared with ACLR alone. Biomechanical studies suggest that LEPs, including ALLR and LET, enhance rotational stability, leading to better clinical outcomes, such as fewer positive pivot shifts and lower graft failure rates^{51,52}. One meta-analysis showed that, overall, ALLR or LET reduced the graft failure risk by approximately 64% and improved rotational stability compared with ACLR alone⁵³. Similarly, Park et al., using NMA, reported that ACLR plus LEP outperformed isolated ACLR in terms of graft failure and positive pivot shift rates¹⁷. This is consistent with our study, which showed that combining LEP with ACLR reduced the graft failure risk and positive pivot shift rates by approximately 70% and 60%, respectively. While prior studies compared different types of LEPs, our analysis focused on graft type, finding that HT and BPTB grafts yielded similar graft failure and pivot shift outcomes when combined with LEP.

The indications for LEP remain debated and inconclusive^{21,54}. LEP is commonly used in revision ACLR, particularly in cases with a high-grade pivot shift, where it effectively reduces failure rates⁵⁵. Its use in primary ACLR is expanding, especially in high-risk patients, such as those who have a high-grade pivot shift, are <25 years old, participate in pivoting

sports, have an elevated tibial slope, or have generalized ligamentous laxity^{56,57}. Sonnery-Cottet et al.⁵⁸ recommended adding LEP to primary ACLR for patients with a chronic ACL injury, Second fracture, or lateral femoral notch sign. Most studies in our analysis (9 of 13) focused on high-risk patients, primarily those with pivot shifts or sports participation^{26,29,30,32,34-38}. However, our meta-regression analysis showed no significant impact of including high-risk patients on outcomes, thereby supporting LEP as a standard adjunct for improving outcomes of primary ACLR regardless of risk status, particularly when using HT and BPTB autografts. This suggests the potential for expanding its indications.

PROMs demonstrated an advantage of augmentation with LEP. In our study, the Lysholm score was significantly higher for HT and BPTB autografts plus LEP than for isolated ACLR (MD: 3 points). However, the observed MD was below the minimal clinically important difference (MCID) of 8.9 for the Lysholm score⁵⁹, and therefore did not reach meaningful clinical impact. Similar patterns have been observed in other meta-analyses, where combined surgery improved subjective IKDC, Lysholm, and Tegner scores but none exceeded the MCID thresholds^{17,53}.

Complications related to graft donor-site morbidity are key concerns in autograft selection for ACLR. BPTB autografts are associated with anterior knee pain, kneeling pain, patellar fracture, patellar tendon rupture, and quadriceps weakness, while HT autografts carry risks of infection and reduced hamstring strength^{46,60}. In our analysis, donor-site morbidity was rarely reported but should still be considered a concern given its importance. Hardware irritation was observed after use of HT autografts, and lateral compartment osteoarthritis was observed with BPTB autografts; however, these complications appeared more related to the LET than to the graft harvesting.

Our study had some limitations. First, as direct comparisons of BPTB and HT autografts were absent, we relied on NMA for indirect comparisons; hence, inconsistency between direct and indirect evidence could not be assessed. Although we performed a transitivity assessment and an additional meta-regression analysis to partially address this limitation, these methods do not fully substitute for formal consistency testing. Second, the limited number of RCTs, particularly those directly comparing BPTB autografts plus LEP to isolated ACLR, resulted in a smaller pool of studies. Third, data on outcomes such as anteroposterior knee stability, various PROMs, strength recovery, and return to sports are crucial in graft selection but could not be analyzed due to study heterogeneity. Fourth, in

each included study, some patients had concomitant injuries or surgeries involving meniscal or cartilage defects in addition to ACLR. These cases have been summarized in Appendix Supplementary Table 1 to aid result interpretation but may have introduced variability into the outcomes. Fifth, variability in inclusion criteria, especially in definitions of high-risk populations and in age criteria across studies, could have introduced bias. Lastly, future research on optimal graft choice should explore alternative grafts, such as the quadriceps tendon and iliotibial band, as viable options.

In conclusion, concurrent ACLR with HT autograft plus LEP was associated with 72%, 73%, and 52% lower odds of graft rupture, graft failure, and clinical failure, respectively, compared with isolated ACLR. Adding LEP to ACLR with BPTB autograft was significantly associated only with a reduction in clinical failure, by approximately 70%, potentially due to the smaller sample size. Both grafts remain viable options for ACLR plus LEP, with the benefit of LEP requiring further validation.

Appendix

 Supporting material provided by the authors is posted with the online version of this article as a data supplement at <http://links.lww.com/JBJS/1828>. ■

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